Tianzhi Xiang

xiangtz@tju.edu.cn · (+86) 13937191026 · Website

Research Interests

Acoustic Metasurfaces • **Wave Mechanics and Control** • **Acoustic–Elastic Wave Coupling**
Acoustic Communication • **Interfacial Wave Transmission**

Education

Tianjin University, School of Mechanical Engineering, Tianjin, China Sep. 2024 – Present

- *M.S.* Candidate in Mechanics | Expected Jun. 2027
- **Core courses**: Mechanical Metamaterials (99/100), Foundation of Solid Mechanics (93/100).

IMDEA Materials Institute, Madrid, Spain Expected Aug. 2026 – Nov. 2026

- *China Scholarship Council (CSC)* funded Visiting *M.S.* Candidate.
- **Host Supervisor:** Prof. Johan Christensen.

Tianjin University, School of Mechanical Engineering, Tianjin, China Sep. 2020 – Jun. 2024

- *B.S.* in Engineering Mechanics (National Strengthening Foundation Plan), GPA: 85/100.
- **Core courses:** Theoretical Mechanics (94/100), Advanced Elasticity (96/100), Vibration and Measurement (90/100), Ordinary Differential Equations and Motion Stability (93/100).

Research Highlights

- First-author manuscript under review on vibro-acoustic metasurfaces for efficient airborne sound transmission through solid barriers, addressing wave coupling across strongly impedance-mismatched interfaces.
- Developed scattering-matrix and interfacial-matching models to describe acoustic–elastic wave transfer, providing a basis for understanding cross-interface coupling and air-coupled elastic-wave excitation.
- Demonstrated metasurface-based platforms for wavefront manipulation, tunable asymmetric transport, and non-contact vibro-acoustic excitation, with a broader interest in extending passive wave-control strategies toward Willis-coupled, programmable, or active acoustic and elastic-wave systems.

Research Experience

[Ongoing] Air-coupled metasurfaces for non-contact elastic-wave excitation and broadband inspection Apr. 2026 – Present

Advisor: Prof. Yanfeng Wang & Prof. Johan Christensen

- Developing metasurfaces for non-contact excitation of elastic waves in solid structures.
- Formulating impedance-based and scattering-matrix models to describe the coupled air–metasurface–cavity–solid interaction and to clarify airborne-to-solid energy transfer mechanisms.
- Investigating asymmetric resonator geometries and modal-hybridization mechanisms, aiming to enhance elastic-wave excitation efficiency and bandwidth.

Vibro-acoustic metasurfaces for sound transmission enhancement through air–solid interfaces
Sep. 2024 – Mar. 2026

Advisor: Prof. Yanfeng Wang

- Developed scattering-matrix and interfacial-matching models to explain airborne-sound-to-elastic-wave conversion across air–solid interfaces.
- Identified metasurface-induced interfacial matching as the key mechanism for overcoming weak sound–structure coupling and enabling thickness-insensitive transmission enhancement.
- Verified high-efficiency airborne sound penetration through solid barriers using acoustic measurements, vibration experiments, and transmission analysis.

Phase–Gradient Acoustic Metasurfaces for Wavefront Manipulation Jul. 2021 – Jun. 2023

Project Leader | Tianjin Municipal Undergraduate Innovation Training Program
Advisor: Prof. Yuesheng Wang & Prof. Yanfeng Wang

- Designed acoustic metasurfaces for wavefront manipulation based on Generalized Snell's Law and impedance descriptions.
- Developed parametric unit cell models, forming a basis for later vibro–acoustic interfacial matching research.

Projects

Reconfigurable Acoustic Moiré Metasurface Oct. 2024 – Jan. 2025

Role: **Second author**.

- Conducted acoustic experiments and analyzed transmission data to characterize tunable asymmetric acoustic transport in three-dimensional free space.
- Evaluated amplitude contrast, propagation direction, and backward suppression under different relative twist configurations.

RoboMaster University League Competition (DJI) Sep. 2022 – Jul. 2023

Role: **Team lead**.

- Led a 40-member multidisciplinary team through one year of preparation, coordinating project planning, technical discussions, task allocation, and competition execution.
- Contributed to system-level R&D planning and mechanical design review, including module integration, milestone tracking, structural feasibility, assembly constraints, and reliability evaluation.
- **Achieved First Prize**.

2024 Global Youth Impact Hackathon (WFUNA) Dec. 2024

Role: Participant.

- Collaborated in a 48-hour multidisciplinary challenge to develop and present a plastic-pollution solution concept under SDG 14, covering feasibility and environmental impact.

Honors & Awards

First-Class Academic Scholarship for Graduate Students	Sep. 2024
Outstanding Graduate of Tianjin University	Jun. 2024
Merit Student of Tianjin University	Dec. 2023
Outstanding Student Leader Award	Sep. 2021

Publications & Patents

- **T. Z. Xiang** et al., "Vibro-acoustic metasurfaces for thickness-independent penetration of airborne sound through solid barriers." Under review at *International Journal of Mechanical Sciences*.
- H. T. Zhou, **T. Z. Xiang** et al., "Reconfigurable Asymmetric Acoustic Transport Enabled by a Reciprocal Moiré Metasurface." Under review at *National Science Review*.
- **T. Z. Xiang** et al., "A Fabrication Method of an Acoustic Metasurface Structure for Efficient Sound Transmission Through Solid Barriers." *Chinese Invention Patent* CN119446109B, granted, 2025.

Technical Skills

- **Language** — Mandarin, English (IELTS Academic 7.0)
- **Software** — COMSOL, SolidWorks, AutoCAD
- **Programming** — MATLAB, Python, LaTeX
- **Computational Design** — Genetic Algorithms, Topology Optimization
- **Experiments** — Acoustic Measurement, Laser Vibrometry